RACE
Regular Analysis Through Continuous Exercise

Sets, Numbers and Intervals : RACE-01

One or more than one correct

1. How many of the following options are correct:

(A) $\{a,b\} \not\subset \{b,c,a\}$

(B) $\{a,e\} \subset \{x : x \text{ is a vowel in the english alphabet}\}$

(C) $\{1,2,3\} \subset \{1,3,5\}$

(D) $\{a\} \subset \{a,b,c\}$

MSN111

2. Given the sets $A = \{1,3,5\}$, $B = \{2,4,6\}$ and $C = \{0,2,4,6,8\}$, which of the following may be considered as universal set (s) for all the three sets A, B and C

(A) {0,1,2,3,4,5,6} (B) ϕ

(C) {0,1,2,3,4,5,6,7,8,9,10} (D) {1,2,3,4,5,6,7,8}

MSN112

Matrix Match Type Question

3. Match the sets in column-I with appropriates roster forms in column-II

Column-I	Column-II
(A) $A = \{x, x \text{ is an odd natural number}\}$	(P) {1,3,5,..}
(B) $B = \{x : x \text{ is an integer}, -1/2 < x < 9/2\}$	(Q) {−2,−1,1,0,2}
(C) $C = \{x : x \text{ is an integer}, x^2 \leq 4\}$	(R) {0,1,2,3,4}
(D) $C = \{x, x \text{ is a letter in the word loyal}\}$	(S) {L,O,Y,A}

MSN113

4. Match the following sets in Roster form in Column-I with set Builder form in Column-II

Column-I	Column-II
(A) {1,2,3,6,9,18}	(P) $\{x : x \text{ is a positive integer and is a divisor of 18}\}$
(B) {1,2,3,6}	(Q) $\{x : x \text{ is a prime number and a divisor of 6}\}$
(C) {2,3}	(R) $\{x : x \text{ is an odd natural number less than 10}\}$
(D) {1,3,5,7,9}	(S) $\{x : x \text{ is natural number and divisor of 6}\}$

MSN114

5.

Column-I (Set)	Column-II (Power Set)	Column-III (Cardinality of power set)
(A) {1}	i. {ϕ,{1},{2},{1,2}}	I. 2
(B) {1,2}	ii. {ϕ,{1},{2},{3},{1,2}, {2,3},{3,1},{1,2,3}}	II. 4
(C) {1,2,3}		III. 8
(D) ϕ	iii. {ϕ,{1}}	IV. 1
	iv. {ϕ}	

MSN115

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Integer Type Question

6. If $A = \{x : x \text{ is a factor of } 42\}$, then write $n(A)$.

MSN116

7. How many of the following statements are true

(i) $\{a, b\} \not\subset \{b, c, a\}$

(ii) $\{a, e\} \subset \{x : x \text{ is a vowel in the english alphabet}\}$

(iii) $\{1,2,3\} \subset \{1,3,5\}$

(iv) $\{a\} \subset \{a, b, c\}$

(v) $\{a\} \in \{a, b, c\}$

(vi) $\{x : x \text{ is an even natural number less than } 6\} \subset \{x : \text{is a natural number which divides } 36\}$

MSN117

8. How many of the following statements are true?

(i) $-2 \in \{-3,0\}$ (ii) $\sqrt{10} \in (3,5)$ (iii) $\sqrt{8} \in (3,5)$ (iv) $\frac{5}{4} \in [0,1]$

(v) $-1.75 \in (-3,2]$ (vi) $2.\overline{9} \in (0,3]$

MSN118

9. Let $A = \{1,2, \{3,4\}\}$. How many of the following statements are incorrect?

(i) $\{3,4\} \subset A$ (ii) $\{3,4\} \in A$ (iii) $\{\{3,4\}\} \subset A$ (iv) $1 \in A$

(v) $1 \subset A$ (vi) $\{1,2,5\} \subset A$ (vii) $\{1,2,3\} \in A$ (viii) $\{1,2,3\} \subset A$

(ix) $\phi \in A$ (x) $\phi \subset A$ (xi) $\{\phi\} \subset A$

MSN119

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ANSWER KEY

Question	1	2	3	4	
Answer	B,D	C	(A→P, B→R, C→Q, D→S)	(A→P; B→S; C→Q; D→R)	

Question	5	6	7	8	9
Answer	(A-iii-I ; B-i-II; C-ii-III, D-iv-IV)	8	3	3	7

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RACE
Regular Analysis Through Continuous Exercise

Sets, Numbers and Intervals : RACE-02

Integer Type Question

1. Taking the set of natural numbers as the universal set, how many of the following are correct?

(i) $\{x : x \text{ is an natural number}\}' = \{x : x \text{ is an odd natural number}\}$

(ii) $\{x : x \text{ is an odd natural number}\}' = \{x : x \text{ is an even natural number}\}$

(iii) $\{x : x \text{ is a positive multiple of 3}\}' = \{x : x \text{ is an odd natural number}\}$

(iv) $\{x : x \text{ is prime number}\}' = \{x : x \text{ is an odd natural number}\}$

(v) $\{x : x \text{ is a natural number divisible by 3 and 5}\}' = \{x : x \text{ is a natural number not divisible by 15}\}$

MSN120

2. If $A = \{5, 10\}$, $B = (5, 10)$, $C = (5, 10]$, $D = [5, 10)$, $E = [5, 10]$, then how many of the following statements are incorrect?

(i) $n(A) + n(B) = 6$ (ii) $B = \{6,7,8,9\}$

(iii) $C = \{6,7,8,9,10\}$ (iv) $n(C) + n(D) + n(E) = 16$

MSN121

3. If $A = \{3,5,7,9,11\}$, $B = \{7,9,11,13\}$, $C = \{11,13,15\}$ and $D = \{15,17\}$; then how many of the following are correct?

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(i) $A \cap B = \{7,9,11\}$ (ii) $B \cap C = \{7,9,11\}$

(iii) $A \cap C \cap D = \{11\}$ (iv) $A \cap C = \{11\}$

(v) $(A \cup D) \cap (B \cup C) = \{\}$

MSN122

4. If $A = \{3,6,9,12,15,18,21\}$, $B = \{4,8,12,16,20\}$, $C = \{2,4,6,8,10,12,14,16\}$, $D = \{5,10,15,20\}$; then how many of the following are correct?

(i) $D - A = \{5,10,20\}$ (ii) $B - C = \{20\}$

(iii) $B - D = \{5,10,15,20\}$ (iv) $C - B = \{2,6,10,14\}$

(v) $D - B = \{5,10,15\}$ (vi) $C - D = \{2,4,6,8,10,12,14,16\}$

(vii) $D - C = \{5,10,15,20\}$

MSN123

Matrix Match Type Question

5. If $U = \{A, B, C, D, E, F, G, H\}$, then match the complements of the following sets :

	Column-I		**Column-II**
(A)	{A,B,C}	(P)	{A,B,C,H}
(B)	{D,E,F,G}	(Q)	{F,B,H,D}
(C)	{A,C,E,G}	(R)	{D,E,F,G,H}
(D)	{F,G,H,A}	(S)	{D,E,C,B}

MSN124

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6. Match the following

	Column-I		**Column-II**
(A)	$[-10,5) \cap (-5,10]$	(P)	$[-5,5]$
(B)	$(-\infty,5] \cap [-5,\infty)$	(Q)	$(-\infty,5)$
(C)	$(-\infty,5] - [5,\infty)$	(R)	$[5,\infty)$
(D)	$(-3,5] \cup [-5,3)$	(S)	$(-5,5)$

MSN125

Subjective Type Question

7. If A and B are any two sets, then verify the following using venn diagram or otherwise :

(i) $A - B = A \cap B'$

MSN126

(ii) $B - A = B \cap A'$

MSN127

(iii) $A - B = A \Leftrightarrow A \cap B = \phi$

MSN128

(iv) $(A - B) \cup B = A \cup B$

MSN129

(v) $(A - B) \cap B = \phi$

MSN130

(vi) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

MSN131

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(vii) $A - (B \cup C) = (A - B) \cap (A - C)$

MSN132

(viii) $A - (B \cap C) = (A - B) \cup (A - C)$

MSN133

ANSWER KEY

Question	1	2	3	4	5	6
Answer	2	4	2	4	(A→R, B→P, C→Q, D→S)	(A→S; B→P; C→Q; D→P)

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RACE
Regular Analysis Through Continuous Exercise

Sets, Numbers and Intervals : RACE-03

Single Correct Question

1. If A and B are two sets, then $A \cap (A \cup B)'$ is equal to-

(A) A (B) B (C) ϕ (D) none of these

MSN134

2. If A is any set, then-

(A) $A \cup A' = \phi$ (B) $A \cup A' = U$ (C) $A \cap A' = U$ (D) none of these

MSN135

3. If A, B be any two sets, then $(A \cup B)'$ is equal to-

(A) $A' \cup B'$ (B) $A' \cap B'$ (C) $A \cap B$ (D) $A \cup B$

MSN136

4. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 5\}$, $B = \{6, 7\}$ then $A \cap B'$ is-

(A) B' (B) A (C) A' (D) B

MSN137

5. If A and B are two sets, then $A \cup B = A \cap B$ iff-

(A) $A \subseteq B$ (B) $B \subseteq A$ (C) $A = B$ (D) none of these

MSN138

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6. Which of the following is a null set?

(A) $\{0\}$ (B) $\{x : x > 0 \text{ or } x < 0\}$

(C) $\{x : x^2 = 4 \text{ or } x = 3\}$ (D) $\{x : x^2 + 1 = 0, x \in \mathbb{R}\}$

MSN139

7. If $Q = \left\{x; x = \frac{1}{y}, \text{ where } y \in \mathbb{N}\right\}$, then-

(A) $0 \in Q$ (B) $1 \in Q$ (C) $2 \in Q$ (D) $\frac{2}{3} \in Q$

MSN140

Multiple Correct Question

8. In a survey it was found that 21 people liked product A, 26 liked product B and 29 liked product C. If 14 people liked products A and B. 12 people liked products C and A, 14 people liked products B and C and 8 liked all the three products. Which of the following are correct?

(A) 11 people like C only

(B) 6 people like B only

(C) 15 people like exactly two products

(D) 20 people like exactly one product

MSN141

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Integer Type Question

9. In a survey of 600 students in a school. 150 students were found to be taking tea and 225 taking coffee.100 were taking both tea and coffee. Find the digit at the tens place of the number of students who were taking neither tea nor coffee?

MSN142

10. In a group of students, 100 students know Hindi, 50 know English and 25 know both. Each of the students knows either Hindi or English. If n is the number of students which are there in the group, then sum of digits of the value of n is-

MSN143

Subjective Type Question

11. Solve the following statements.

(P) If X and Y are two sets such that X has 40 elements, $X \cup Y$ has 60 elements and $X \cap Y$ has 10 elements, how many elements does Y have?

(Q) In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many people like both coffee and tea?

(R) In a committee, 50 people speak French, 20 speak Spanish and 10 speak both Spanish and French. How many speak at least one of these two languages?

(S) Out of 500 car owners investigated, 400 owned car A and 200 owned car B, 50 owned both A and B cars. Is this data correct?

MSN144

12. How many of the following statements are correct?

(P) If $X \cup Y$ has 50 elements, X has 28 elements and Y has 32 elements, then $X \cap Y$ has 10 elements.

(Q) In a school there are 20 teachers who teach mathematics or physics. If 12 teach mathematics and 4 teach both physics and mathematics, then 12 teachers take physics.

(R) In a class of 35 students, 24 like to play cricket and 16 like to play football. If each students likes to play at least one of the two games, then 5 students like to play both cricket and football.

(S) In a survey of 400 students in a school, 100 were listed as taking apple juice, 150 as taking orange juice and 75 were listed as taking both apple as well as orange juice. Then 225 students were taking neither apple juice nor orange juice.

MSN145

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	B	B	C	D	B	A,B,D	2	8
Question	11					12				
Answer	(P-30, Q-19, R-60, S-No.)					4				

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Regular Analysis Through Continuous Exercise

Sets, Numbers and Intervals : RACE-04

Single Correct Questions:

1. If $n \in I$, then set of all integers can be denoted by

(A) $\{2n, 2n + 1\}, n \in I$ (B) $\{3n, 3n + 1, 3n + 2\}, n \in I$

(C) $\{4n, 4n + 1, 4n + 2, 4n + 3\}, n \in I$ (D) All of the above

MSN152

2. If $x.y = 36$ and $x \in \mathbb{N}$; $y \in \mathbb{N}$, then number of ordered pairs of (x, y) is given by -

(A) 7 (B) 8 (C) 9 (D) 10

MSN153

3. If $x, y \in \mathbb{N}$, then number of ordered pairs such that $\frac{1}{x}+\frac{1}{y}=1$ is

(A) 0 (B) 1 (C) 2 (D) 3

MSN154

4. How many integers in from 500 to 2020 (both inclusive) are multiple of 3 or 7?

(A) 725 (B) 652 (C) 73 (D) 651

MSN155

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5. Consider the number N = 7 7 7 4 9 5 8 P 9 6 Q. If N is divisible by 4, then find the number of possible ordered pairs (P, Q).

(A) 25 (B) 30 (C) 40 (D) 50

MSN156

6. Consider the number N = 7 8 7 4 9 5 8 P 9 6 Q.

If N is divisible by 8 and 9 both, then find the number of possible ordered pairs (P, Q).

(A) 1 (B) 2 (C) 3 (D) 4

MSN157

Subjective Type Question

7. If $a\sqrt{2}+b=3\sqrt{2}+4$, find the integral value of a, b and justify your answer.

MSN158

8. Express the following in form of p/q, where $p, q \in \mathbb{Z}$ and $q \neq 0$

(i) $0.\overline{18}$

MSN159

(ii) $0.\overline{16}$

MSN160

(iii) $0.4\overline{23}$

MSN161

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9. $\sqrt{a}+\sqrt{b}$ and $a - b$ are rational numbers. Prove that $\sqrt{a}, \sqrt{b}$, a and b all are rational.

MSN162

10. The last digit of a six-digit positive integer is 6. If 6 is moved to the front as the first digit, the resulting number is 4 times the original positive integer. Find the smallest possible value of the original positive integer.

MSN163

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6
Answer	D	C	B	D	B	C

7. $a = 3, b = 4$ **8.** (i) $\frac{2}{11}$ (ii) $\frac{16}{99}$ (iii) $\frac{419}{990}$ **10.** 153846

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